

MARCO ANTONIO GONZALEZ JUNIOR

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Solutions Architect | Cloud Engineer | Artificial Intelligence Specialist

PROFESSIONAL SUMMARY

Solutions Architect with 16+ years building scalable enterprise systems and 10+ years in AI/ML. Led technical teams delivering solutions that generated 30% revenue increases and 200% improvement in customer retention across financial, retail, and digital marketing. Currently specializing in advanced generative AI: custom model development, fine-tuning, RAG, agentic workflows, multi-agent systems, and enterprise AI governance.

SKILLS

Software Architecture: Domain-Driven Design, Clean Architecture, Event-Driven Architecture, Microservices, Enterprise Integration Patterns

Solutions Architecture: Cloud-Native Design, API Strategy, System Integration, Scalability Patterns, Cost Optimization, Vendor Evaluation, Technology Roadmaps

Cloud & DevOps: AWS/Azure/GCP, Kubernetes, Docker, Terraform, CI/CD, Infrastructure as Code, Serverless

AI & Machine Learning: Generative AI, LLMs, Fine-tuning, RAG Systems, Knowledge Graphs, Multi-Agent Systems, MLOps, LLMOps

Technical Leadership: Hands-on Leadership, Technology Strategy, Innovation, Evangelism, Digital Transformation

EXPERIENCE

Sabesp

Lead AI Specialist

São Paulo

September 2025 – Present

Leading AI teams on innovative projects, driving Generative AI enablement and digital transformation throughout the company.

- Collaborating with AI/ML scientists to research, design, and deploy generative AI solutions addressing real-world operational challenges across water and sanitation infrastructure.
- Interacting with internal customers to understand business problems, guide them on adoption patterns, and establish paths to production for generative AI solutions.
- Implementing MLOps/LLMOps CI/CD pipelines on Databricks and Azure, serving foundation and fine-tuned models powering specialized agentic systems for customer support and code review.
- Architecting centralized AI Gateway providing model access with fallbacks, spend tracking, rate limiting, security, and prompt caching.

Wipro

Senior Solutions Architect – Artificial Intelligence

São Paulo

September 2024 – September 2025

Led AI-driven legacy modernization initiatives using knowledge graphs and LLMs to transform enterprise COBOL systems into modern technology stacks.

- Designed intelligent legacy modernization platform using Neo4j knowledge graphs, LangGraph, and AWS Bedrock to automate COBOL application transformation, delivering 85%+ reduction in migration timelines.
- Engineered production RAG systems combining semantic search (Pinecone, Weaviate) and knowledge graphs, achieving 90% fidelity in preserving business logic during COBOL to Java/.NET migration.
- Led team of 8 engineers developing intelligent AI workflows, enabling automated extraction of functional requirements from enterprise legacy systems – reducing manual documentation time by 90% across millions of lines of code.

IBM

Senior Solutions Architect – Artificial Intelligence

São Paulo

July 2022 – September 2024

Architected advanced AI solutions for financial services clients, including foundation model fine-tuning and hybrid RAG systems with continuous learning capabilities.

- Fine-tuned custom LLMs specialized to business domains through fine-tuning pipelines and RAG knowledge bases using PyTorch and TensorFlow, achieving 75% automated resolution rate and 25% improvement in customer satisfaction.
- Designed multi-cloud MLOps platform (AWS, Azure) with automated continuous learning pipelines, ensuring 99% uptime while serving 10M+ monthly interactions.
- Developed comprehensive model evaluation framework integrating BERTScore semantic analysis and human-in-the-loop feedback, enabling continuous model refinement under strict security and compliance constraints.
- Standardized AI model deployments through Docker, Kubernetes and Terraform, cutting deployment times by 80% and ensuring reliable, repeatable release cycles.

banQi

Senior Solutions Architect

São Paulo

June 2019 – July 2022

Led technical architecture and engineering teams at fast-growing fintech startup, promoted from Technical Lead to Principal Engineer, driving system scalability and strategic technical direction.

- Architected microservices platform using Docker, Kubernetes, TypeScript and Go supporting 10x traffic growth, reducing system response times by 65% and achieving 99.95% uptime.
- Built ML-powered fraud detection system with real-time scoring, reducing fraudulent transactions by 40% and false positives by 25%.
- Unified system architecture implementing Domain-Driven Design and serverless computing (AWS Lambda, Node.js, TypeScript), establishing technical standards aligned with business strategy.
- Led adoption of containerization across 15+ applications using Helm, ArgoCD, and GitOps workflows, reducing deployment times by 80%.
- Built and managed cross-functional engineering team of 12+ developers, implementing agile practices with GitHub Actions CI/CD and increasing delivery velocity by 45%.

Petlove

Tech Lead

São Paulo

May 2015 – June 2019

Led technical architecture for Brazil's largest pet supplies e-commerce platform, driving digital transformation initiatives.

- Architected prime membership platform using Ruby on Rails, Elixir, and PostgreSQL, achieving 30% of monthly revenue (\$2M+) and 500% increase in customer repurchase rate.
- Led ERP system modernization using Python and distributed warehouse management, enabling multi-state expansion and supporting 200% business growth.
- Built Business Intelligence team of 6 engineers, implementing big data and machine learning solutions with ETL pipelines that improved inventory forecasting accuracy by 85%.
- Designed automated supply chain optimization using machine learning algorithms, reducing procurement costs by 15% and eliminating 90% of stockout incidents.

Jüssi

Solutions Architect

São Paulo

November 2012 – May 2015

Architected digital marketing automation solutions as Facebook Marketing Partner and Google Partner for enterprise clients.

- Led Facebook Marketing Partner certification, collaborating with Facebook Engineering using GraphQL APIs to deliver marketing automation for biggest e-commerces in Brazil.
- Architected Google Ads automation platforms using Java and RESTful APIs, managing \$5M+ monthly ad spend and achieving 40% improvement in ROAS.
- Built real-time bidding optimization algorithms using statistical modeling, reducing customer acquisition costs by 25% across 5k+ enterprise campaigns.
- Designed scalable attribution platform using message queues and distributed processing, processing 10M+ daily events with sub-second response times.

Editora Abril

Senior Software Engineer

São Paulo

September 2008 – November 2012

Developed large-scale content management systems for Brazil's leading media and entertainment publisher.

- Built Veja SP platform using HTML5, CSS3, JavaScript, and MySQL, serving 500K+ monthly users with 99.9% uptime.
- Architected enterprise content management system using Java and SOA, supporting 20+ publications and 200+ concurrent editors.
- Optimized database performance using advanced SQL queries and caching strategies, reducing page load times by 60% and supporting 10x traffic growth.

PAST EXPERIENCES

Serasa Experian (Java Enterprise Architect), Western Union (Senior Software Engineer), Toyota do Brasil (Software Engineer), Wdotnet Internet (Owner), Flyever Electronics (Software Engineer)

EDUCATION

Faculdade de Informática e Administração Paulista

Faculdade Impacta de Tecnologia

Universidade Federal de São Carlos

MBA Data Science (2015–2016)

BSc Information Systems (2011–2015)

BSc Materials Engineering (1996–2000)

CERTIFICATIONS

AWs: Solutions Architect Professional (2025), Data Engineer Associate (2025), ML Engineer (2024), ML Specialty (2024), AI Practitioner (2024), Solutions Architect Associate (2022), Developer (2022)

Specializations: Stanford Machine Learning and Deep Learning (2024), IBM AI Engineering Professional (2024), Data Science (2015), Recommender Systems (2014)

LANGUAGES

Portuguese: Native

English: Fluent

Spanish: Conversational